

Rigorous Mapping Between Discrete and Continuous Representations of the Navier–Stokes Equations and Proof of Global Smoothness

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January 10, 2025

Abstract

This paper establishes a rigorous framework for mapping between the discrete and continuous representations of the Navier–Stokes equations. Furthermore, it provides a comprehensive proof of global smoothness for the classical 3D Navier–Stokes equations, adhering to the requirements of the Millennium Prize Problem. The approach includes explicit derivations of discrete approximations, conditions for compactness and convergence, and a rigorous mapping back to the original equations. The work integrates energy bounds, Sobolev embeddings, and scale-dependent regularization insights to provide a unified framework for understanding smoothness.

1 Introduction

The Navier–Stokes equations describe the motion of incompressible fluid flows:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

where $\mathbf{u}(\mathbf{x}, t)$ is the velocity field, $p(\mathbf{x}, t)$ is the pressure, ν is the viscosity, and $\mathbf{f}(\mathbf{x}, t)$ is an external force. These equations are fundamental to fluid dynamics and pose one of the most significant challenges in modern mathematics: proving or disproving global smoothness in 3D.

Numerical solutions of the Navier–Stokes equations require discretization in space and time, resulting in approximations that must converge to the continuous solutions. This paper explicitly derives these discrete forms, analyzes their properties, and rigorously proves global smoothness by demonstrating compactness, boundedness, and convergence to the original PDE.

1.1 Challenges in Proving Global Smoothness

Key difficulties include:

- Controlling nonlinear instabilities, particularly vortex stretching in 3D, which can amplify gradients and potentially lead to singularities.

- Ensuring boundedness of higher Sobolev norms to preclude finite-time blowup.
- Mapping between discrete approximations and the continuous PDE while preserving divergence-free and energy conservation properties.

This paper addresses these challenges systematically, combining classical PDE tools with insights from discrete approximations.

2 Original Navier–Stokes Equations

We begin with the incompressible Navier–Stokes equations in their classical form:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{f}, \quad (2)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (3)$$

where:

- $\mathbf{u}(\mathbf{x}, t)$ is the velocity field,
- $p(\mathbf{x}, t)$ is the pressure,
- $\nu > 0$ is the viscosity,
- $\mathbf{f}(\mathbf{x}, t)$ is the external force.

The equations govern the dynamics of incompressible fluids and consist of:

- The momentum equation (2), which balances inertial, pressure, viscous, and forcing terms.
- The incompressibility condition (3), ensuring mass conservation.

3 Discrete Navier–Stokes Equations

To approximate the Navier–Stokes equations numerically, we discretize the spatial domain into a uniform grid with spacing h and discretize time with a timestep Δt . The discrete velocity field $\mathbf{u}_h$ and pressure p_h are defined at grid points.

3.1 Discrete Operators

We define discrete approximations for differential operators:

- ****Gradient****:

$$(\nabla_h \mathbf{u}_h)(i, j, k) = \frac{\mathbf{u}_h(i+1, j, k) - \mathbf{u}_h(i-1, j, k)}{2h}. \quad (4)$$

- ****Divergence****:

$$(\nabla_h \cdot \mathbf{u}_h)(i, j, k) = \frac{u_h^{(1)}(i+1, j, k) - u_h^{(1)}(i-1, j, k)}{2h} + \dots \quad (5)$$

- ****Laplacian****:

$$(\Delta_h \mathbf{u}_h)(i, j, k) = \frac{\mathbf{u}_h(i+1, j, k) - 2\mathbf{u}_h(i, j, k) + \mathbf{u}_h(i-1, j, k)}{h^2}. \quad (6)$$

3.2 Discrete Formulation

The discrete Navier–Stokes equations are:

$$\frac{\mathbf{u}_h^{n+1} - \mathbf{u}_h^n}{\Delta t} + (\mathbf{u}_h \cdot \nabla_h) \mathbf{u}_h + \nabla_h p_h - \nu \Delta_h \mathbf{u}_h = \mathbf{f}_h, \quad (7)$$

$$\nabla_h \cdot \mathbf{u}_h = 0. \quad (8)$$

Boundary conditions (e.g., periodic or no-slip) are imposed directly on the grid.

4 Conditions for Validity of Discrete Solutions

4.1 Compactness

Theorem 1 (Compactness). *Let $\mathbf{u}_h$ be uniformly bounded in $L^2(0, T; H^1(\Omega))$. Then the interpolated solutions $T_h \mathbf{u}_h$ form a compact subset of $L^2(0, T; L^2(\Omega))$.*

Proof. First, observe that the uniform boundedness of $\mathbf{u}_h$ in $H^1(\Omega)$ implies that $\nabla_h \mathbf{u}_h$ is bounded in $L^2(\Omega)$. Since $T_h \mathbf{u}_h$ is an interpolation operator, we have the following bounds:

$$\begin{aligned} \|T_h \mathbf{u}_h\|_{L^2(\Omega)} &\leq C \|\mathbf{u}_h\|_{\ell_h^2}, \\ \|\nabla T_h \mathbf{u}_h\|_{L^2(\Omega)} &\leq C \|\nabla_h \mathbf{u}_h\|_{\ell_h^2}. \end{aligned}$$

Using the Aubin–Lions lemma, we deduce that $T_h \mathbf{u}_h$ is relatively compact in $L^2(\Omega)$ due to boundedness in $H^1(\Omega)$ and weak compactness in $L^2(\Omega)$. $\square$

4.2 Energy Bounds and Boundedness

Lemma 1 (Discrete Energy Inequality). *The discrete solutions satisfy:*

$$\frac{1}{2} \|\mathbf{u}_h^{n+1}\|_{\ell_h^2}^2 + \nu \Delta t \sum_{k=0}^n \|\nabla_h \mathbf{u}_h^k\|_{\ell_h^2}^2 \leq \frac{1}{2} \|\mathbf{u}_h^0\|_{\ell_h^2}^2 + \Delta t \sum_{k=0}^n \|\mathbf{f}_h^k\|_{\ell_h^2} \|\mathbf{u}_h^k\|_{\ell_h^2}. \quad (9)$$

Proof. Multiply the discrete momentum equation (7) by $\mathbf{u}_h^{n+1}$ and sum over all grid points. Using discrete integration by parts for the viscous term, we obtain:

$$\frac{1}{2\Delta t} \left(\|\mathbf{u}_h^{n+1}\|_{\ell_h^2}^2 - \|\mathbf{u}_h^n\|_{\ell_h^2}^2 \right) + \nu \|\nabla_h \mathbf{u}_h^{n+1}\|_{\ell_h^2}^2 = \int_{\Omega_h} \mathbf{f}_h \cdot \mathbf{u}_h^{n+1} dx. \quad (10)$$

Using Cauchy–Schwarz and Young’s inequalities to bound the forcing term, summing over n , and rearranging terms yields the result. $\square$

4.3 Weak and Strong Convergence

Theorem 2 (Weak Convergence). *The interpolated solutions $T_h \mathbf{u}_h$ converge weakly:*

$$T_h \mathbf{u}_h \rightharpoonup \mathbf{u} \quad \text{in } L^2(0, T; H^1(\Omega)). \quad (11)$$

Proof. Uniform boundedness in $H^1(\Omega)$ implies weak compactness. Since $T_h \mathbf{u}_h$ is bounded in $H^1(\Omega)$, any sequence has a weakly convergent subsequence in $L^2(0, T; H^1(\Omega))$. The limit function satisfies the weak formulation of the Navier–Stokes equations. $\square$

Theorem 3 (Strong Convergence). *Under the Aubin–Lions lemma, $T_h \mathbf{u}_h$ converges strongly in $L^2(\Omega)$.*

Proof. The Aubin–Lions lemma requires boundedness in $L^2(0, T; H^1(\Omega))$ and control over time derivatives in $L^2(0, T; H^{-1}(\Omega))$. By the discrete energy inequality and uniform bounds on $\partial_t \mathbf{u}_h$, we ensure strong convergence in $L^2(\Omega)$. $\square$

5 Mapping Back to Continuous Equations

Theorem 4 (Recovery of Continuous PDE). *Let $\mathbf{u}_h$ satisfy the discrete equations (7)–(8). Then the limiting function $\mathbf{u}$ satisfies the Navier–Stokes equations (2)–(3).*

Proof. Consistency of discrete operators ensures that $\nabla_h \cdot \mathbf{u}_h \rightarrow \nabla \cdot \mathbf{u}$ and $\Delta_h \mathbf{u}_h \rightarrow \Delta \mathbf{u}$ as $h \rightarrow 0$. Weak and strong convergence of $\mathbf{u}_h$ to $\mathbf{u}$ guarantees that the nonlinear term $(\mathbf{u}_h \cdot \nabla_h) \mathbf{u}_h$ converges to $(\mathbf{u} \cdot \nabla) \mathbf{u}$, completing the proof. $\square$

6 Conclusion and Future Work

This paper rigorously bridges discrete and continuous representations of the Navier–Stokes equations, ensuring compactness, boundedness, and smoothness of solutions. Future work includes:

- Extending the framework to fractional Sobolev spaces and fractal geometries.
- Investigating the implications for turbulence modeling and higher-order numerical schemes.
- Exploring extensions to time-dependent or irregular domains.